Math 142, F19 **Name:** Answer
Quiz 2

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (4 points) Evaluate the integral $\int \frac{1}{e^{-x}+e^{x}}\,dx$

$$\int \frac{1}{e^{-x}+e^{x}}dx = \int \frac{1}{e^{-x}+e^{x}}\cdot\frac{e^{x}}{e^{x}}dx = \int \frac{e^{x}}{1+(e^{x})^2}dx$$

let $u=e^{x}$, $du=e^{x}dx$

Then

$$\int \frac{1}{e^{-x}+e^{x}}dx = \int \frac{1}{1+u^2}du$$
$$= \arctan(u)+C$$
$$= \arctan(e^{x})+C$$

Problem 2. (6 points) Evaluate the following integrals using integration by parts:

(a) $\int \ln(2x)\,dx$

Let $u=\ln(2x), dv=dx$

$\Rightarrow du=\frac{1}{2x}\cdot 2dx=\frac{1}{x}dx, v=x$

$$\int \ln(2x)dx = x\ln(2x)-\int x\cdot\frac{1}{x}dx$$
$$= x\ln(2x)-\int dx$$
$$= x\ln(2x)-x+C$$

(b) $\int_0^1 te^{-t}\,dt$

Let $u=t, dv=e^{-t}dt$

$du=dt, v=-e^{-t}$

Thus

$$\int_0^1 te^{-t}dt = t(-e^{-t})\Big|_0^1 - \int_0^1(-e^{-t})dt$$
$$= [-e^{-1}-0]+\int_0^1 e^{-t}dt$$
$$= -e^{-1}+[-e^{-t}]_0^1$$
$$= -e^{-1}+[-e^{-1}-(-1)]$$
$$= -\frac{2}{e}+1$$